GRATIA: A Mobile-Native Blockchain Protocol

Whitepaper v0.5

"Freedom and Gratitude Through Universal Participation"

Executive Summary

Gratia is a Layer 1 blockchain where the phone is the miner and your daily life is the proof of work.

The problem: Bitcoin requires ASICs. Ethereum requires $80,000+ in staked capital. Solana requires datacenter hardware. Five billion people own smartphones, but no blockchain lets them participate. The industry that promised decentralization built a new class of gatekeepers.

The solution: Gratia's consensus mechanism is architecturally dependent on mobile hardware — GPS, accelerometers, Bluetooth, NFC, secure enclaves — that servers and mining rigs simply do not have. This creates a hardware moat: only real phones held by real people can mine. No amount of capital can bypass it.

How it works:

- **Proof of Life** runs silently in the background. Normal phone usage — unlocking, carrying, charging, connecting to Wi-Fi — generates cryptographic attestation that you are a real human. Zero battery impact. Zero user action.
- **Proof of Presence** activates when the phone is plugged in above 80% battery. Mining earns GRAT at a flat rate. Every minute pays the same. No diminishing returns.
- **Three-pillar security** requires Proof of Life + staking + energy expenditure simultaneously. Phone farms get caught by behavioral analysis. Emulators fail

sensor attestation. Whales are capped by staking overflow pools that redistribute to small miners.

Governance: One phone, one vote. Not token-weighted. A user with 1 GRAT has the same governance power as a user with 1 million.

Performance: 4-second block time, 128 TPS base throughput scaling to 2,560 TPS with geographic sharding, 256 KB blocks, Bulletproofs zero-knowledge attestations, WASM smart contracts with mobile-native opcodes.

Current status: The protocol is running on real phones. Two Android devices — a $90 Samsung A06 and a flagship S25 — produce blocks, send verified transactions, discover peers automatically, and reach consensus over the internet. The Rust core spans 13 crates with 705 automated tests and zero warnings. The GratiaScript smart contract compiler is operational — contracts compile to WASM and execute in a pure-Rust interpreter with mobile-native opcodes no other blockchain has. This is not a whitepaper for a product that might be built. The product exists.

Token: GRAT (smallest unit: 1 Lux = 0.000001 GRAT). Fair launch — 85-90% emitted through mining, no private presale at a discount, 25% annual emission reduction.

Abstract

Gratia is a mobile-native layer-1 blockchain and smart contract platform designed from the ground up to run exclusively on smartphones. By leveraging the full hardware capabilities of modern mobile devices — GPS, NFC, accelerometers, secure enclaves, cellular radios, Bluetooth, cameras, barometers, and more — Gratia creates a consensus mechanism that is impossible to replicate on servers, ASICs, or desktop hardware. The result is a truly decentralized network where the unit of participation is one phone, one person, restoring the original promise of cryptocurrency: a peer-to-peer system accessible to everyone.

Bitcoin's whitepaper envisioned "one CPU, one vote." In practice, industrial mining has concentrated power among a small number of entities with access to capital-intensive hardware. Gratia corrects this by anchoring consensus to the most widely distributed computing device on earth — the smartphone — owned by over 5 billion people worldwide.

The protocol operates on a dual-proof system: **Proof of Life** — passive background attestation that accumulates as users live their normal lives — and **Proof of Presence** — active mining that engages when the phone is plugged in and charged above 80%. Consensus security is reinforced by a three-pillar architecture combining Proof of Life, staking, and energy expenditure, where all three must be satisfied simultaneously. Governance follows a one-phone-one-vote model, ensuring that no amount of wealth can override the collective voice of the network's participants.

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1. The Problem

1.1 The Broken Promise of Decentralization

Bitcoin was designed to be a peer-to-peer electronic cash system where anyone could participate in securing the network using a personal computer. Today, Bitcoin mining is dominated by industrial operations running purpose-built ASIC hardware in warehouses, consuming more energy than many countries. The average person has been effectively locked out of meaningful participation.

Ethereum transitioned to Proof of Stake, but meaningful validator participation requires 32 ETH (~$80,000+ at typical prices), creating a financial barrier that excludes the vast majority of the world's population.

Even "lightweight" blockchains like Solana require high-performance server hardware to run validator nodes. The blockchain industry has replicated the same centralization it was designed to eliminate — replacing banks with mining farms and staking cartels.

1.2 The Unbanked Paradox

Approximately 1.4 billion adults worldwide remain unbanked, yet over 5 billion people own smartphones. The technology to participate in a global financial system is already in their hands — but no existing blockchain is designed to run on it natively. Gratia closes this gap.

1.3 The Energy Narrative

Bitcoin mining consumes an estimated 150+ TWh annually. Governments worldwide are enacting legislation to restrict or ban proof-of-work mining due to environmental

concerns. Gratia's approach uses idle capacity on devices that are already manufactured, already owned, and already drawing power from chargers. The marginal energy cost of participation is negligible compared to any existing consensus mechanism.

1.4 The Competitive Landscape: Why Existing Mobile Projects Have Failed

Gratia is not the first project to attempt mobile cryptocurrency mining. Understanding why previous attempts have fallen short is essential to understanding why Gratia is fundamentally different.

Pi Network has attracted over 35 million users with a mobile mining app. However, Pi's "mining" consists of tapping a button once every 24 hours. The phone performs no actual computational work, validates no transactions, and secures no network. Pi is effectively a loyalty points program with cryptocurrency branding. The phones do not participate in consensus in any meaningful way. Gratia's phones actually secure the network — they produce blocks, validate transactions, execute smart contracts, and perform real computational work using real electricity. The distinction is between a phone that mines and a phone that pretends to mine.

Electroneum launched a mobile mining feature but ultimately shifted to a centralized validation model where mining rewards were distributed based on engagement rather than actual consensus participation. The phone was a wallet with a rewards drip, not a network participant.

Phonon, MobileCoin, and other mobile-focused projects have generally built mobile wallets on top of conventional blockchain architectures. The phone is the interface, not the infrastructure. The actual consensus runs on servers.

What makes Gratia fundamentally different is that the phone is not an accessory to the network — it IS the network. The consensus mechanism cannot run on servers because it requires GPS, accelerometers, Bluetooth, NFC, and the full sensor stack of a real smartphone. No previous project has made the phone itself the irreplaceable unit of consensus. This is not a technical feature — it is an architectural decision that determines who can participate and who cannot. In Gratia, servers cannot participate. Only phones can. Only people can.

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2. Design Philosophy

2.1 Core Principles

Phone-native, not phone-compatible. Gratia is not a mobile wallet for a desktop blockchain. The protocol itself is architecturally dependent on mobile hardware capabilities. Nodes that cannot prove they are real phones in real locations held by real people cannot participate in consensus.

One phone, one voice. The network is designed so that owning more phones does not linearly scale influence. Diminishing returns, geographic distribution requirements, and multi-sensor attestation make Sybil attacks a physical logistics problem, not a computational one. Once a node crosses the consensus threshold, it has equal voice in the network — regardless of the phone's price, model, or available sensors beyond the minimum.

Your life is your proof. The protocol does not ask users to perform any special action. Normal phone usage — unlocking, carrying, connecting to Wi-Fi, charging — passively generates the attestation data required to participate. Gratia observes what you already do and recognizes it as proof that you are a real person with a real phone. If you use your phone like 98% of humans use their phones, you qualify.

Plug in to mine. Active mining (block production and validation) engages only when two conditions are met: the phone is connected to a power source, and the battery is charged above 80%. This can happen at any time of day — overnight charging, at a desk during work, on a couch in the evening. The trigger is the state of the phone, not the time of day.

No single sensor is mandatory. The attestation system is a composite score where no individual hardware component is required. A phone missing NFC, a barometer, or a SIM card can still reach the consensus threshold by compensating through other available sensors. The threshold is deliberately set so that any smartphone manufactured after 2018 with a working GPS, accelerometer, and either Wi-Fi or Bluetooth can cross it.

Accessible by design. The consensus algorithm targets the hardware profile of mid-range smartphones (e.g., Snapdragon 600-series, MediaTek Dimensity 700-series, or equivalent), ensuring that participation does not require flagship devices. If a phone

costs $50+ and was manufactured in the last 7 years, it should be able to participate fully.

2.2 The Hardware Moat

Gratia's competitive advantage is that its consensus mechanism requires capabilities that servers and desktop computers do not possess. By weaving GPS, NFC, accelerometers, barometers, ambient sensors, and secure enclaves into the protocol, Gratia creates a network that can only exist because smartphones exist. This is not a limitation — it is a moat.

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3. The Dual-Proof System

Gratia operates on two complementary proof mechanisms that mirror the natural rhythm of phone usage.

3.1 Proof of Life (Passive Mode)

When: Whenever the phone is NOT meeting mining parameters (unplugged, or plugged in but below 80% battery). Also runs continuously in the background regardless of mining state, as daily Proof of Life must be maintained every day to remain eligible for mining.

What: The Gratia app runs silently in the background, collecting attestation data from the phone's normal operation. This process has zero noticeable impact on battery life, performance, or user experience. The user never needs to open the app for Proof of Life to accumulate.

How it works: Proof of Life observes the natural patterns of daily phone usage and packages them into a cryptographic attestation that proves the device is a real phone, used by a real human, in a real location. The user installs the app, plugs in their phone, and begins mining immediately — no waiting period. From that point forward, the node must maintain valid Proof of Life every day to keep mining eligibility. The privilege to mine is granted instantly and maintained through ongoing honest participation.

Proof of Life is the primary defense against phone farms, emulators, and all forms of fake participation. A phone sitting in a rack in a warehouse — no matter how sophisticated the software running on it — cannot produce a valid Proof of Life because it is not being carried, used, moved through the world, and interacted with by a real human. The specific parameters that constitute a valid Proof of Life are defined in Section 4.

3.2 Proof of Presence (Mining Mode)

When: Whenever BOTH of the following conditions are met:

1. The phone is connected to a power source (wall charger, car charger, battery pack, solar panel — any external power)
2. The battery level is at or above 80%

Additionally, the node must have a valid Proof of Life for the current day and must have staked the minimum required GRAT.

These conditions can be met at any time of day. A phone plugged in at a desk during work hours mines. A phone on a car charger during a commute mines (once it hits 80%). A phone charging overnight mines. The trigger is the state, not the time.

What: The phone becomes an active participant in the Gratia blockchain consensus. It produces and validates blocks, processes transactions, executes smart contracts, and earns GRAT rewards.

How it works:

Activation sequence. When the phone detects a power connection, it checks the battery level. If below 80%, the phone prioritizes charging — the user's need for a charged phone always comes first. Once the battery crosses 80%, the app verifies that a valid Proof of Life exists for the current day and that the minimum stake is in place, then enters Mining Mode.

Flat reward rate. Mining rewards are earned at a flat rate for every minute the phone is actively mining. There are no diminishing returns, no tiers, no multipliers. Whether you mine for 2 hours at a coffee shop or 14 hours overnight, every minute earns the same reward. Honest participation is compensated equally regardless of session length.

Thermal and battery management. The mining algorithm continuously monitors CPU temperature and battery state. If the CPU approaches thermal limits, workload is throttled. If the battery drops below 80% (which can happen if the charger's output is lower than the mining power draw on very slow chargers), mining pauses until it recovers above 80%. The protocol is designed to never degrade the user's phone experience or damage their device.

Deactivation. When the phone is unplugged, or if the battery falls below 80% and doesn't recover within a grace period, the node exits Mining Mode and continues Proof of Life passive collection. The transition is seamless and invisible to the user.

3.3 The Daily Rhythm

The dual-proof system maps to the natural rhythm of human phone usage:

Morning: User unplugs phone. This unplug event registers as part of today's Proof of Life charge cycle requirement. Mining Mode deactivates. Proof of Life continues passively collecting attestation data.

Throughout the day: User carries phone, uses it normally, connects to various Wi-Fi networks, interacts with the screen, encounters different Bluetooth environments. Proof of Life accumulates. If the user plugs in at any point (at work, in a car, at a cafe) and the phone crosses 80%, Mining Mode activates and earns GRAT until they unplug. When they unplug, it seamlessly returns to passive Proof of Life collection.

Evening/Night: User plugs in phone. Battery charges to 80%. Mining Mode activates. The phone mines through the night, earning GRAT while the user sleeps.

The cycle repeats every day. The user never opens the Gratia app. They never perform any special action. They simply live their life, and the network rewards them for being a real person with a real phone.

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4. Proof of Life: Parameters

4.1 Design Principle

The Proof of Life parameters are calibrated so that 98% of smartphone users pass simply by using their phone the way they already use it. No special actions, no

behavioral changes, no awareness of the requirements is needed. The parameters are designed to be invisible to honest users and impossible for phone farms to satisfy at scale.

4.2 Required Daily Parameters

A valid Proof of Life for a given day requires ALL of the following to be satisfied within a rolling 24-hour window:

Human Presence — Proving a real person is using this phone:

- **Minimum 10 unlock events** spread across the day. The global average is 80-100 unlocks per day, so this threshold captures only phones that are essentially never touched. The unlocks must be distributed across at least a 6-hour window (not 10 unlocks in 2 minutes).
- **Screen interaction events** showing organic touch patterns at multiple points throughout the day. The protocol records only the timing and frequency of interactions — never the content of what is being viewed, typed, or tapped. Human interaction patterns are irregular and varied; automated patterns are detectable by their regularity and mechanical consistency.
- **At least one orientation change** — the phone was physically picked up, rotated, tilted, or moved between positions (e.g., flat on a table to upright in a hand). This is detected by the accelerometer and/or gyroscope. A phone that maintains a single orientation for 24 hours is not being used by a human.

Physical Movement — Proving the phone exists in the real world:

- **Accelerometer data showing human-consistent motion** during at least a portion of the day. This does not require vigorous activity — the subtle vibrations of a phone resting on a table in a lived-in environment, the shift of being picked up and set down, or the bounce of being carried in a pocket all qualify. A bedridden person who picks up their phone a few times still passes. A phone in a static rack does not.
- **At least one GPS fix** confirming a plausible geographic location. The fix does not need to be high-precision — Wi-Fi-assisted positioning with 100-meter accuracy is sufficient. The location must be consistent with previous days (no teleportation between distant cities within impossible timeframes).

Network Environment — Proving the phone is in a real, changing environment:

- **Connection to at least one Wi-Fi network OR detection of Bluetooth peers** at some point during the day. This confirms the phone is operating in a real wireless environment.
- **Varying Bluetooth peer environments** at some point during the day. The phone must detect meaningfully different sets of nearby Bluetooth devices at different times — such as the difference between a home environment and a workplace, or the changing Bluetooth landscape of a commute. A phone that sees the exact same set of Bluetooth devices 24 hours a day is in a static, artificial environment.

Usage Cycle — Proving physical human interaction with the device:

- **At least one charge cycle event** — either a plug-in or unplug event — during the 24-hour period. This proves that a human physically connected or disconnected a cable from the device. A phone that remains plugged in continuously without interruption for multiple days fails this parameter. This is the simplest and most elegant proof of human interaction: someone picked up a cable and put it in the phone, or took it out.

4.3 What Passes Proof of Life

The following real-world scenarios all produce a valid Proof of Life:

- **A typical working adult:** Unplugs phone in the morning. Carries it to work. Uses it throughout the day. Plugs in at night. Easily passes every parameter.
- **A stay-at-home parent:** Phone moves between rooms throughout the day. Encounters varying Bluetooth environments (neighbors, visitors, smart home devices). Regular screen interactions. Plug/unplug cycle. Passes.
- **An elderly person with minimal phone use:** Picks up phone a few times per day to check messages or make calls. Phone sits on a nightstand or table otherwise. 10+ unlocks is achievable for anyone who uses their phone at all. Movement is minimal but present. Plug/unplug cycle from daily charging. Passes.
- **A student:** Heavy phone usage throughout the day. Multiple Wi-Fi networks (home, school, library). Dynamic Bluetooth environment. Easy pass.
- **A bedridden or disabled person:** Can still unlock their phone 10+ times. Orientation changes occur when the phone is picked up from a bedside table. Bluetooth environment changes as caregivers, visitors, and other household members move in and out. Plug/unplug cycle from daily charging. Passes.

- **A person in a developing country with a $50 phone on Wi-Fi only:** No SIM card, no cellular data. GPS works via Wi-Fi-assisted positioning. Wi-Fi connection from home or a community access point. Phone is unlocked and used regularly. Plug/unplug from solar panel or shared charger. Passes.

4.4 What Fails Proof of Life

The following scenarios produce an invalid Proof of Life:

- **A phone farm:** 500 phones sitting in racks. Zero unlock events (nobody is unlocking 500 phones organically). Flat accelerometer data (no human motion). Same Wi-Fi BSSID 24/7 (no network transitions). Same Bluetooth peer set 24/7 (every phone sees the same 499 other phones permanently). No charge cycle events (all phones plugged in continuously). Fails on EVERY parameter simultaneously.
- **An emulator or virtual machine:** No real accelerometer data (or synthetic data that lacks the micro-noise characteristics of real hardware). No real GPS hardware (spoofed coordinates lack satellite metadata). No real Bluetooth radio (cannot discover actual nearby devices). No charge cycle events (software cannot simulate physical cable interaction). Fails.
- **A phone left in a drawer unused:** Zero unlock events. Zero screen interactions. Zero orientation changes. No Bluetooth environment variation. Potentially no charge cycle if it's left to die. Fails — and appropriately so, because an unused phone is not contributing to network security.

4.5 Grace Period

Mining begins immediately upon installation — there is no onboarding delay. However, the privilege must be maintained. If a node fails to produce a valid Proof of Life on a single day, mining eligibility is preserved but flagged. If Proof of Life is invalid for two consecutive days, mining eligibility is paused. Eligibility resumes immediately upon the next valid Proof of Life day — no extended re-onboarding is required. This "mine now, keep the privilege" model means honest users are never waiting to earn, while fraudulent nodes are quickly detected and removed.

4.6 Privacy Architecture

All Proof of Life data is processed entirely on-device. Raw sensor data — GPS coordinates, accelerometer readings, Wi-Fi network names, Bluetooth device

identifiers, screen interaction logs — NEVER leaves the phone. The protocol uses zero-knowledge proofs to submit a cryptographic attestation to the network that states: "This device produced a valid Proof of Life today: YES/NO." The network can verify the attestation's validity without learning any of the underlying data. The user's location, behavior patterns, daily routine, and device identity are private by design.

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5. Three-Pillar Consensus Security

5.1 Architecture

Gratia's consensus security rests on three pillars that must ALL be satisfied simultaneously for a node to participate in block production and validation. Each pillar defends against a different category of attack, and together they form a redundant defense that covers essentially every realistic threat vector.

Pillar 1: Proof of Life — The Primary Wall Stops phone farms and mass Sybil attacks. A phone that is not used by a real human in a real, changing environment cannot participate. This is the most powerful defense in the protocol because it cannot be defeated with money or computing power — it requires a real human life.

Pillar 2: Staking — The Economic Commitment Layer Stops small-scale multi-device gaming. Even if someone manages to fake Proof of Life on a few devices (by hiring people to carry phones, for example), they must also commit economic stake to each device. The capped staking model (see Section 6.3) ensures that splitting capital across multiple devices yields diminishing returns compared to honest single-device participation.

Pillar 3: Energy Expenditure — The Physical Resource Layer Stops virtual and emulated nodes. The phone must be physically plugged into a power source and performing real computational work on a real ARM chipset. This is a physical constraint that cloud servers and emulators cannot satisfy — they can simulate sensor data, but they cannot simulate being a real phone drawing real power from a real charger.

5.2 Why All Three Are Required

Proof of Life alone could theoretically be defeated by hiring humans to carry phones and use them normally. Those phones would produce valid Proof of Life attestations because they ARE being used by real humans. But with staking, each of those phones must also have economic commitment — splitting capital across hired carriers is expensive and yields diminishing returns via the overflow pool mechanism. And with energy expenditure, each phone must be plugged in and above 80% to mine — coordinating charging schedules for hired carriers adds operational complexity.

Staking alone is just Proof of Stake — a system where money equals power. Whales dominate. This is exactly what Gratia exists to prevent.

Energy expenditure alone is just Proof of Work — a system where hardware and electricity equal power. Mining farms dominate. Also what Gratia exists to prevent.

Any two without the third leaves a meaningful attack vector open. All three together create a system where the only efficient path to earning GRAT is to be a real person, with a real phone, with genuine economic commitment, using real electricity. In other words: honest participation.

5.3 Progressive Slashing

If a node is caught acting dishonestly — submitting invalid blocks, fabricating attestations, attempting to double-spend — it faces graduated consequences designed to protect honest users from catastrophic loss while making sustained cheating economically devastating.

Offense Escalation (90-Day Rolling Window):

Offense	Stake Impact	Mining Impact
1st offense	None	48-hour mining pause
2nd within 90 days	10% of effective stake burned	Mining resumes after slash
3rd within 90 days	50% of effective stake burned	30-day mining lockout
Proven fraud (any time)	100% burned permanently	Permanent ban

The 90-day rolling window means a single bad day doesn't haunt a node forever. After 90 days of clean participation, the offense count resets. But proven fraud — where validator committee consensus confirms deliberate cheating — bypasses the escalation ladder entirely and results in immediate full slash.

Slash Distribution:

- For proven fraud: 70% of slashed stake is burned (deflationary), 30% is distributed to the validator committee members who confirmed the fraud
- For all other offenses: 100% burned
- Reporter cap: no single validator can earn more than their own stake from a single fraud report, preventing false-flagging incentives

Multi-Pillar Consequences: Beyond the economic penalty, slashed nodes also face consequences across all three pillars:

- **Proof of Life history reset** — the node must accumulate a fresh day of valid Proof of Life before mining can resume
- **Device attestation flagged** — the device's secure enclave identity is marked, making it harder to rejoin with the same physical device
- **Behavioral scrutiny increased** — slashed nodes face enhanced PoL verification for 90 days after reinstatement

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6. Tokenomics

6.1 The GRAT Token

Ticker: GRAT **Maximum supply:** [TBD — to be determined based on emission modeling] **Smallest unit:** 1 Lux (1 GRAT = 1,000,000 Lux) — named for light, reflecting accessibility and illumination.

6.2 Emission and Distribution

Fair launch model. Gratia follows a fair launch approach inspired by Bitcoin's original distribution:

Genesis Block. The first block is mined by the founding team on real phones, under the same rules every future participant will follow. The founders are users first.

Founding allocation: 10-15% of total supply, with strict constraints:

- **Development Fund** (locked, vesting over 4 years): Funds engineering, security audits, and protocol development. Spending is transparent and reported to the

community.

- **Founding Team** (locked for 1 year, then vesting over 3 years): The team cannot sell tokens in the first year and can only gradually access them over the following three years. This ensures the team is aligned with long-term network health, not short-term profit.
- **Ecosystem Grants Fund**: Funds developer tools, educational resources, localization, and community-driven projects that expand the Gratia ecosystem.

No private investor pre-sale at a discount. Gratia will not sell tokens to venture capital firms or private investors at prices below what the public can access. If external funding is needed during development, it will come through equity investment in the company building the protocol (not discounted tokens), public token sales at uniform pricing, or crowdfunding mechanisms like Kickstarter.

The remaining 85-90% of tokens are emitted exclusively through mining — earned by real people on real phones. No shortcuts, no backdoors. The emission schedule is fixed in the protocol from day one and cannot be changed without governance approval.

Early miner advantage. Just as Bitcoin's early miners earned more per block when the network was small, Gratia's early miners will earn more per session because fewer participants share the rewards. This is fair because early miners take the biggest risk — participating in an unproven network with tokens of uncertain value. As the network grows and more phones join, the per-node reward naturally decreases but the token's value presumably increases with adoption.

Emission schedule.

- Mining rewards are distributed to all nodes actively participating in Proof of Presence mining. Rewards are earned at a flat rate per minute of active mining — no diminishing returns, no tiers, no multipliers.
- Geographic equity: Nodes in underserved regions (fewer existing nodes per population) earn elevated rewards, incentivizing network growth where it matters most.
- Halving: Emission rate reduces by 25% annually (gentler than Bitcoin's 50% halving to maintain mining incentives for small participants over a longer period).

6.3 Staking Model: Capped with Overflow Pool

Gratia uses a capped staking model that turns wealth concentration into a benefit for the entire network rather than a tool for dominance.

Minimum stake: A minimum amount of GRAT must be staked to participate in mining. This serves as an anti-spam mechanism and ensures every mining node has something to lose if they act dishonestly. The minimum is set low enough to be accessible to any participant and may be adjusted by governance as the token's value changes.

Per-node stake cap: Each node has a maximum effective stake — for example, 1,000 GRAT. Staking up to this cap contributes directly to the node's economic commitment and is subject to slashing if the node acts dishonestly.

Overflow to Network Security Pool: Any GRAT staked above the per-node cap automatically flows into the Network Security Pool. The whale still earns yield on their full staked amount, but their excess capital beyond the cap is redistributed as additional rewards to ALL active mining nodes proportionally. This means:

- A whale staking 100,000 GRAT contributes 1,000 to their own node and 99,000 to the pool that benefits everyone.
- Whales are incentivized to stake because they earn yield on the full amount.
- Individual consensus power is capped regardless of wealth.
- The richer the whales, the more the small miners benefit.
- Wealth concentration is structurally converted from a centralizing force into an equalizing one.

Minimum Stake Calculation:

The minimum stake is pegged to approximately 14 days of average mining rewards at the current network size, calculated as:

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min_stake = (daily_budget / active_miners) × 14
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This self-adjusting formula ensures the stake is always "two weeks of your own output" — cheap enough when the network is young to not block new participants, but meaningful enough when the network is mature to serve as a real deterrent. A phone farm operator running 100 phones must lock up 100× this amount, turning scale into a liability rather than an advantage.

The per-node stake cap is set at 100× the minimum stake and scales automatically with it.

6.4 Fee Structure

- **Transaction fees:** Minimal, paid in GRAT. Fees are burned (deflationary pressure).
- **Smart contract gas:** Denominated in Lux, calculated based on ARM compute cycles consumed rather than abstract "gas units," making costs intuitive and predictable.
- **NFC transactions:** Zero-fee for tap-to-pay transfers under a threshold amount (e.g., 10 GRAT), encouraging everyday use as digital cash.

6.5 Intrinsic Token Value

Unlike many cryptocurrencies whose value is derived purely from speculation, GRAT has an intrinsic floor value anchored in physical reality: the energy cost to produce it.

Every GRAT token is created through real computational work performed by a real phone consuming real electricity from a real power source. The energy consumed by a phone's ARM CPU during Mining Mode — measured in kilowatt-hours — represents an irreducible cost of production. No rational miner would sell GRAT below what it cost them in electricity to mine it. This creates a natural price floor from day one, before any exchange listing, before any speculative market forms.

This is the same fundamental argument that gives Bitcoin a floor value — the cost of mining provides a baseline. But Gratia's energy-backed value is arguably more defensible because the production cost is distributed across millions of individual devices worldwide rather than concentrated in a handful of industrial facilities. There is no single entity that can flood the market with below-cost tokens because no single entity controls a meaningful share of production.

As the network grows and more phones participate, the total energy expended to secure the network increases, reinforcing the intrinsic value of each token. GRAT is, at its most fundamental level, a commodity — a digital commodity whose creation requires measurable, verifiable expenditure of a real-world resource.

6.6 Token Utility

GRAT is designed for active use, not passive holding. The token serves multiple functions within the Gratia ecosystem:

Peer-to-peer transactions. GRAT functions as digital cash for everyday payments between individuals, with NFC tap-to-pay enabling instant in-person transfers. Zero-fee transactions below a threshold amount make micro-payments viable for daily commerce.

Smart contract execution. GRAT is used to pay gas fees for deploying and interacting with smart contracts on the GratiaVM. Location-triggered contracts, proximity contracts, presence contracts, and environmental oracle contracts all require GRAT for execution.

Staking. GRAT is staked to participate in mining and earn rewards. The capped staking model with overflow pool creates ongoing demand for staking while ensuring excess capital benefits the entire network.

On-chain polling and verified human consensus. This is a utility unique to Gratia and potentially one of its most valuable applications. Any GRAT holder can create an on-chain poll that is open to all wallets on the network. Because every wallet is backed by a Proof-of-Life-verified human, every response is guaranteed to come from a unique, real person. This creates the world's first incorruptible polling infrastructure.

The implications extend far beyond the crypto ecosystem. Every online poll today — Twitter polls, Reddit votes, change.org petitions, product surveys — can be trivially manipulated by bots and fake accounts. Gratia's on-chain polling produces results that are verifiably human, one-person-one-vote, and tamper-proof. Use cases include:

- **Protocol governance:** Voting on network upgrades, parameter changes, and community proposals.
- **Political polling:** Gathering genuine public opinion on political questions, verified to be one real human per vote, immune to bot manipulation.
- **Market research:** Companies paying to access verified human responses rather than bot-contaminated survey data.
- **Community decision-making:** DAOs, cooperatives, homeowner associations, clubs, and any group that needs trustworthy collective decisions.
- **Dispute resolution:** On-chain arbitration where verified human jurors render decisions.

Organizations would pay GRAT to create polls and access the verified human response infrastructure, creating organic token demand driven by real-world utility

rather than speculation. The cost to create a poll (denominated in GRAT) is burned, providing deflationary pressure proportional to the platform's utility.

Governance participation. GRAT staking is a prerequisite for voting in protocol governance, ensuring voters have economic commitment to the network's health.

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7. Governance: One Phone, One Vote

7.1 Principle

Gratia's governance is democratic at the device level. Voting power is not proportional to token holdings — it is one phone, one vote. A farmer in Indonesia with 50 GRAT staked has the same governance vote as a fund manager with 500,000 GRAT. Governance reflects the will of the people using the network, not the people who accumulated the most wealth on the network.

7.2 Proposal Eligibility

Any node with 90 or more consecutive days of valid Proof of Life history can submit a governance proposal. This prevents spam proposals from brand-new accounts while keeping the bar achievable for anyone who has been a genuine, consistent participant in the network.

7.3 Governance Process

Discussion period: 14 days. The proposal is visible in the app to all nodes. Users can read the proposal, review community discussion, and form their opinion. This gives the community time to evaluate complex technical or economic changes rather than rushing a vote.

Voting period: 7 days. Every node that currently maintains a valid Proof of Life and has staked the minimum gets exactly one vote — yes, no, or abstain. The vote is cast directly in the app with a simple tap. One phone, one vote.

Passage threshold: 51% of votes cast. A simple majority of participating voters is required to pass a proposal.

Quorum requirement: 20% of all active mining nodes. At least 20% of all nodes currently eligible for mining must participate in the vote for it to be valid. This prevents a tiny minority from pushing through changes when the broader community is not paying attention.

Implementation delay: 30 days. After passage, there is a 30-day window before the change takes effect. This gives node operators, developers, and the ecosystem time to prepare for the change. If a critical flaw is discovered during this window, an emergency reversal vote can be triggered with a lower proposal threshold.

7.4 Emergency Governance

For critical security patches that cannot wait for the full 51-day governance cycle (14 + 7 + 30), a supermajority of 75% of a randomly selected validator committee (selected by the same weighted VRF used for block production) can fast-track a security fix. This emergency mechanism can ONLY be used for security-related changes — not economic parameters, not governance rules, not philosophical changes. Any emergency fix must be ratified by a standard governance vote within 90 days or it automatically reverts.

7.5 Self-Amending Protocol

The governance system itself is subject to governance. The community may, through the standard proposal process, modify voting thresholds, quorum requirements, discussion periods, or any other governance parameter. The protocol is opinionated at launch (one phone, one vote) but humble enough to let its participants evolve it through the mechanisms it provides.

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8. Mobile Hardware Utilization Map

The following details how every major sensor and hardware component in a modern smartphone is leveraged by the Gratia protocol. Components are categorized as **Core** (required for consensus threshold), **Standard** (common, boosts Presence Score), or **Enhanced** (less common or opt-in, further boosts score).

8.1 ARM Processor (CPU) — Consensus Engine [CORE]

Role: Executes the core consensus algorithm and smart contract virtual machine.

Design: Gratia uses a custom consensus algorithm optimized for ARM architecture instruction sets. The algorithm leverages ARM-specific features including NEON SIMD instructions and hardware cryptographic accelerators (AES, SHA) present in all modern ARM SoCs. The algorithm is designed to run efficiently within the thermal and power constraints of mobile chipsets while being computationally inefficient on x86/x64 server architectures.

Anti-ASIC/Server Strategy: The algorithm's computational core rotates through operations that specifically exploit the heterogeneous big.LITTLE core architecture found in mobile ARM chips. Server CPUs with uniform core designs cannot efficiently context-switch between the algorithm's phases, creating a natural performance penalty for non-mobile hardware.

8.2 GPU (Mobile) — Parallel Validation [CORE]

Role: Handles parallelized transaction verification and smart contract execution.

Design: Mobile GPUs (Adreno, Mali, PowerVR) have fundamentally different architectures than desktop/server GPUs (NVIDIA, AMD). Gratia's transaction validation pipeline is designed as a mobile GPU compute shader workload, using OpenCL ES or Vulkan Compute profiles that are native to mobile GPUs. The workload is tuned to the memory bandwidth and core count typical of mobile GPUs (4-12 cores) rather than desktop GPUs (thousands of cores), meaning desktop GPUs gain no meaningful advantage.

8.3 GPS — Location Verification [CORE]

Role: Geographic location attestation and Sybil resistance.

Design: Every participating node must periodically submit a location attestation derived from GPS data. The protocol requires:

- At least one GPS fix per day for Proof of Life validity
- Consistency between daily fixes (no teleportation)
- Wi-Fi-assisted positioning is sufficient — high-precision GPS is not required
- Geographic diversity bonus: nodes in underserved regions earn elevated mining rewards, incentivizing global distribution organically

GPS data is processed on-device and submitted as a zero-knowledge attestation. The network verifies that the node is in a valid, consistent location without learning the actual coordinates.

8.4 Accelerometer — Human Behavior Verification [CORE]

Role: Detecting that a device is in the physical possession of a real human, not mounted in a rack.

Design: The accelerometer is the backbone of Proof of Life's humanity verification. It feeds a Human Interaction Model that analyzes motion patterns passively:

- **Micro-movements:** A phone held by a human or resting on a surface in a lived-in environment exhibits characteristic micro-vibrations that differ from a phone in a static rack.
- **Daily motion signature:** Over a 24-hour cycle, a phone carried by a human shows movement during waking hours and relative stillness during sleep. A phone in a rack shows near-zero variance with no daily rhythm.
- **Orientation changes:** A phone that is picked up, set down, flipped, or rotated is being physically handled by a person.

The protocol does not require vigorous activity — it requires evidence of a phone existing in a human's daily life.

8.5 Wi-Fi Radio — Network Connectivity and Location Breadcrumbs [CORE]

Role: Primary network connectivity for Wi-Fi-only devices and passive location attestation.

Design: The Wi-Fi radio serves as both a connectivity layer and an attestation source:

- **Connectivity:** For phones without active SIM cards, Wi-Fi is the primary connection to the Gratia network. Wi-Fi-only phones are full participants with no penalty.
- **BSSID breadcrumbs:** As the phone connects to different Wi-Fi networks throughout the day (home, work, public spaces), each network's unique hardware address creates a passive location breadcrumb trail without requiring GPS.

- **Environmental attestation:** The set of visible Wi-Fi networks in a given area creates a location fingerprint that cross-references GPS data.

8.6 Bluetooth — Peer Discovery, Mesh, and Sybil Detection [CORE]

Role: Peer discovery, offline mesh networking, local consensus clusters, and phone farm detection.

Design: Bluetooth Low Energy (BLE) enables several critical functions:

- **Proof of Life parameter:** Varying Bluetooth peer environments throughout the day are a required Proof of Life parameter. A phone that sees the same set of devices 24/7 is in a static, artificial environment.
- **Peer discovery:** Nodes automatically discover nearby Gratia participants via BLE advertisements, forming local clusters that can share transaction data and perform preliminary validation.
- **Mesh relay:** In areas with poor cellular or Wi-Fi coverage, Bluetooth mesh networking allows transactions to hop between nearby phones until one with connectivity can broadcast to the main network.
- **Sybil detection:** BLE scanning reveals how many Gratia nodes are within physical proximity. An implausible density of nodes in a small area is flagged.

8.7 Cellular Radio (LTE/5G) — Optional Network Backbone [STANDARD]

Role: Network connectivity and supplementary location attestation for phones with active SIM cards.

Design: The cellular radio is NOT required — phones without SIM cards or cellular plans are full participants via Wi-Fi. For phones that do have cellular connectivity, the radio provides:

- **Supplementary connectivity:** Block propagation and transaction broadcast over the cellular network, especially useful when Wi-Fi is unavailable.
- **Cell tower fingerprint:** The unique combination of serving and neighboring cell tower IDs, signal strengths, and timing advance values creates a location fingerprint that is extremely difficult to spoof. This supplements GPS-based location attestation and increases the node's Presence Score.

- **Network-level timestamp:** Cellular networks provide highly accurate time synchronization, supplementing NTP-based timing.

8.8 Barometer — Altitude Verification and Weather Oracle [STANDARD]

Role: Location verification and native weather data oracle.

Design: The barometer provides atmospheric pressure readings that serve dual purposes:

- **Altitude cross-reference:** Barometric pressure correlates with altitude, providing a third axis to GPS verification.
- **Weather data oracle:** The network of millions of barometric sensors becomes a decentralized weather data oracle. Aggregated, anonymized pressure readings from Gratia nodes could provide real-time atmospheric data with unprecedented granularity — a potential revenue stream for the protocol and a real-world utility no other blockchain can provide.

8.9 Magnetometer — Environmental Fingerprinting [STANDARD]

Role: Secondary location verification and environmental uniqueness.

Design: The magnetometer measures the local magnetic field, which varies based on geographic location, nearby structures, and even the specific room a phone is in. This creates an Environmental Magnetic Signature that:

- Provides a secondary cross-reference for location attestations
- Detects when multiple phones claim different locations but share identical magnetic environments (phone farm in the same room)
- Adds entropy to the node's unique identity fingerprint

8.10 NFC (Near Field Communication) — Tap-to-Transact [STANDARD]

Role: Zero-intermediary peer-to-peer transactions and proximity-based contract execution.

Design: NFC enables the most intuitive transaction experience possible:

- **Tap to pay:** Two phones tap together to initiate a direct peer-to-peer token transfer with no internet required at the moment of transaction. The transaction is queued locally and broadcast to the network when connectivity is available.
- **Proximity-verified contracts:** Smart contracts that require physical proximity — proof that two parties are in the same location.
- **Trust handshake:** Two nodes that NFC-tap establish a Proximity Trust Bond — a cryptographic attestation that two unique physical devices were in the same location at the same time.

8.11 Ambient Light Sensor — Time-of-Day Verification [STANDARD]

Role: Cross-referencing claimed timezone and time-of-day consistency.

Design: A node claiming to be in a specific timezone should report ambient light levels consistent with the time of day. This provides time-of-day plausibility checks, indoor/outdoor detection, and additional entropy for environmental fingerprinting.

8.12 Fingerprint Sensor / Face ID — Biometric Transaction Authorization [STANDARD]

Role: Securing transaction signing and wallet access.

Design: The biometric sensor provides transaction signing authorization for high-value transactions and periodic biometric check-ins that confirm the device owner is present. Biometric data is processed entirely within the secure enclave — no biometric information is ever transmitted.

8.13 Secure Enclave / TEE (Trusted Execution Environment) — Cryptographic Core [STANDARD]

Role: The trust anchor for cryptographic operations.

Design: The secure enclave (Apple Secure Element, ARM TrustZone, Android StrongBox) provides:

- **Key generation and storage:** Private keys are generated within and never leave the secure enclave.
- **Hardware attestation:** The secure enclave can produce a cryptographic proof that it is running on genuine hardware (not an emulator or virtual machine).

- **Sensor data signing:** All sensor attestations are signed within the secure enclave before transmission, preventing tampering.
- **Device uniqueness:** Each secure enclave has a unique hardware identity burned in during manufacturing, providing a strong guarantee of device uniqueness.

8.14 Camera — Environmental Verification [ENHANCED, OPT-IN]

Role: Optional environmental attestation (privacy-preserving).

Design: Camera usage is strictly optional and privacy-first. If the user opts in, the camera captures a single frame of the environment, computes a perceptual hash locally, and immediately discards the image. This hash provides a unique environmental fingerprint without transmitting any identifiable imagery. No images are ever stored or transmitted.

8.15 Microphone — Ambient Audio Fingerprinting [ENHANCED, OPT-IN]

Role: Optional environmental uniqueness verification (privacy-preserving).

Design: If the user opts in, the microphone captures a brief ambient audio sample (1-2 seconds), computes an acoustic fingerprint locally, and immediately discards the raw audio. The fingerprint is non-reversible and cannot reconstruct the original audio. It provides environmentally unique and temporally unique attestation data. Only the cryptographic hash of the acoustic features is transmitted.

8.16 Wi-Fi Direct — Offline Transaction Layer [STANDARD]

Role: High-bandwidth offline transactions and data sync between nearby nodes.

Design: Wi-Fi Direct enables phone-to-phone data transfer at Wi-Fi speeds without requiring a router or internet connection:

- **Offline transaction queuing:** Users in areas without data can conduct transactions via Wi-Fi Direct, with broadcast when connectivity is regained.
- **Block propagation in low-connectivity areas:** Nodes sync blockchain state with nearby connected nodes, reducing bandwidth costs.
- **Emergency resilience:** In disaster scenarios where cellular infrastructure fails, the network continues processing local transactions via Wi-Fi Direct mesh.

8.17 Battery and Charging State — Mining Mode Controller [CORE]

Role: Determining when a node is eligible for mining.

Design: The battery and charging subsystem enforces the two hard rules of mining:

- **Rule 1: Plugged in.** The phone must detect an active power connection to participate in consensus.
- **Rule 2: Battery above 80%.** Mining does not activate until the phone reaches 80% charge. The user's need for a charged phone always comes first.
- **Charge cycle event:** At least one plug-in or unplug event per day is a required Proof of Life parameter.
- **Thermal management:** The mining algorithm monitors CPU temperature and throttles workload to prevent overheating, preserving device longevity.

8.18 Storage (Flash Memory) — Lightweight Chain Architecture [CORE]

Role: Blockchain state storage within mobile storage constraints.

Design: Mobile devices have limited storage, requiring an aggressive approach:

- **Pruned state:** Nodes store only the current state and recent block headers, not the full transaction history.
- **Sharded storage:** Blockchain state is sharded across geographic regions. Each node primarily stores the shard relevant to its geographic area.
- **Target footprint:** Maximum 2-5 GB, ensuring participation is viable on budget devices with 32 GB total storage.
- **Compression:** Aggressive compression optimized for flash storage read/write patterns.

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9. Consensus Threshold: Who Gets In

9.1 Design Constraint

The consensus threshold is designed so that at least 50% of smartphones currently in active use worldwide can cross it. This means the threshold must be achievable by

phones that cost as little as $50, were manufactured as early as 2018, may not have NFC, barometer, magnetometer, or fingerprint sensors, may not have an active SIM card or cellular plan, and may operate exclusively on Wi-Fi.

9.2 Core Requirements (Must Have All)

To cross the consensus threshold, a phone must demonstrate four fundamental capabilities present in virtually every smartphone manufactured since 2015:

1. **ARM Processor** — must pass a lightweight computational challenge confirming a real ARM mobile chipset.
2. **GPS** — must produce a location fix (Wi-Fi-assisted accuracy is sufficient).
3. **Accelerometer** — must demonstrate human-consistent motion patterns.
4. **Wi-Fi OR Bluetooth** — must have at least one for network connectivity and peer discovery.

9.3 Binary Threshold, Equal Voice

The consensus threshold is binary — pass or fail. There are no tiers, no levels, no earning multipliers based on phone quality. A $50 phone on Wi-Fi only that crosses the threshold earns the same base mining reward per minute as a $1,200 flagship on 5G. Once you're in, you're equal.

9.4 Composite Presence Score (Above Threshold)

While the threshold is binary, nodes above it still generate a composite Presence Score based on all available sensor data (ranging from 40 to 100). This score does NOT affect mining rewards. It affects only the probability of being selected for block production and validation committee membership — a security function that ensures nodes with richer attestation data take on the most critical protocol responsibilities. But they earn no more for doing so.

Sensor / Capability	Score Bonus	Approx. Penetration
Core four (required)	40 (base)	~95% of smartphones
Gyroscope data	+5	~80%
Ambient light sensor	+3	~80%
Bluetooth peer discovery (in addition to Wi-Fi)	+5	~85%

Active cellular connection + cell tower data	+8	~70%
Barometer / altitude data	+5	~60%
Magnetometer / compass	+4	~65%
NFC capability	+5	~60%
Secure enclave / hardware attestation	+8	~55%
Fingerprint or biometric sensor	+5	~60%
Camera-based environment hash (opt-in)	+4	~95% (opt-in)
Microphone ambient fingerprint (opt-in)	+4	~95% (opt-in)
Participation history (30+ days)	+2	Earned over time
Participation history (90+ days)	+2 additional	Earned over time

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10. Smart Contract Platform: GratiaVM

10.1 Design Constraints

The Gratia Virtual Machine (GratiaVM) is purpose-built for mobile execution:

- **ARM-optimized bytecode:** GratiaVM compiles to a custom bytecode set optimized for ARM NEON instructions, achieving near-native performance on mobile chipsets.
- **Strict resource limits:** Contracts have hard limits on memory (256 MB), execution time (500ms), and storage writes per transaction, reflecting the reality of running on battery-powered devices.
- **Mobile-native capabilities:** Unlike the EVM, GratiaVM has native opcodes for location queries, NFC triggers, proximity verification, and time-of-day logic — capabilities that are impossible on server-based blockchains.

10.2 Mobile-Native Smart Contract Capabilities

Gratia's smart contracts can do things no other blockchain can because the execution environment has access to mobile hardware:

Location-Triggered Contracts: Payments or actions that execute when a party physically arrives at a specified location. Use cases: delivery confirmation, gig economy arrival verification, tourism rewards, geofenced access control.

Proximity Contracts: Contracts that require two or more parties to be physically co-located (proven via NFC tap or Bluetooth detection). Use cases: in-person commerce escrow, physical asset handoffs, meeting verification.

Presence Contracts: Contracts that require proof of sustained physical presence over a time period. Use cases: event attendance (conferences, concerts), work shift verification, community service logging.

Environmental Oracle Contracts: Contracts that trigger based on real-world environmental data gathered from the sensor network. Use cases: parametric weather insurance (using barometer data from local nodes), air quality bonds, agricultural monitoring.

Time-Zone-Aware Contracts: Contracts that natively understand the physical timezone of each party (verified by ambient light, GPS, and cellular data) rather than relying on UTC. Use cases: international commerce with timezone-appropriate deadlines, local market hours enforcement.

10.3 Developer Experience

- **Language:** GratiaScript — a TypeScript-derived language with mobile-native primitives (`@location`, `@proximity`, `@presence`, `@sensor`) that compile to GratiaVM bytecode. Familiar syntax lowers the barrier for the millions of existing TypeScript/JavaScript developers.
- **Testing:** Developers test contracts on their own phones using the Gratia DevKit app. No testnet servers required — the phone IS the development environment.
- **Deployment:** Contracts are deployed from the phone, maintaining the ethos that everything in the ecosystem is phone-first.

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11. Network Architecture

11.1 Layer Structure

- **Layer 0 — Mesh Layer:** Bluetooth and Wi-Fi Direct mesh for offline resilience and local peer discovery.
- **Layer 1 — Consensus Layer:** The core Proof of Presence blockchain running on cellular/Wi-Fi connectivity.
- **Layer 2 — Application Layer:** Smart contracts, dApps, and developer tools.
- **Layer 3 — Oracle Layer:** Decentralized sensor data aggregation (weather, atmospheric, environmental) from the node network.

11.2 Node Types

Node Type	Hardware	Connectivity	Role
Mining Node	Phone, plugged in, above 80% battery	Cellular/Wi-Fi	Block production, validation, consensus
Proof of Life Node	Phone, unplugged or below 80%	Cellular/Wi-Fi	Wallet, transaction sending/receiving, Proof of Life accumulation
Mesh Node	Phone (any state)	Bluetooth/Wi-Fi Direct	Offline transaction relay, peer discovery
Archive Node	Server (non-consensus)	Broadband	Historical data storage, block explorer, API

Critical distinction: Archive nodes cannot participate in consensus. They serve the ecosystem by providing historical data access, but they have no governance power and earn no mining rewards. Consensus is exclusively for phones.

11.3 Block Structure

Blocks are designed for mobile constraints:

- **Target block size:** 256 KB (small enough for reliable propagation over cellular networks)
- **Block time target:** 3-5 seconds
- **Block header:** Includes merkle roots for transactions, state, sensor attestations, and geographic distribution metrics
- **Propagation:** Optimized for mobile network latency with compact block relay (only transaction hashes, not full transactions, for nodes that already have the

transactions in their mempool)

11.4 Block Production

Block production follows a Weighted Random Selection model:

1. All nodes currently in Mining Mode with valid Proof of Life and staked minimum are eligible.
2. A VRF (Verifiable Random Function) selects a block producer, weighted by Presence Score. Higher-scoring nodes are more likely to be selected for block production, but all eligible nodes earn the same flat mining reward regardless.
3. The selected producer assembles the block and broadcasts it.
4. A committee of 21 validators (also selected by weighted VRF from different geographic regions) validates and signs the block.
5. Finality is achieved when 14/21 validators sign (67% threshold).

11.5 Scalability and Throughput

Base layer throughput. With 256 KB blocks (262,144 bytes) and an average transaction size of approximately 400 bytes (larger than Bitcoin's ~250 bytes due to attestation metadata), each block accommodates roughly 655 transactions. At the target block time range:

- At 3-second block times: approximately 218 transactions per second (TPS)
- At 5-second block times: approximately 131 transactions per second (TPS)

For context: Bitcoin handles approximately 7 TPS. Ethereum handles approximately 15-30 TPS. Visa handles roughly 1,700 TPS on average with peaks of 24,000 TPS. Gratia's base layer throughput of 131-218 TPS is significantly faster than Bitcoin and Ethereum but below centralized payment networks.

Why this is sufficient for early growth. At 131-218 TPS, the network can process 11-19 million transactions per day. For a network of even 1 million active nodes, this provides substantial capacity for peer-to-peer payments, smart contract interactions, and governance voting.

Scaling to 10 million+ nodes. As the network grows beyond base layer capacity, Gratia employs geographic sharding — the blockchain is partitioned into regional shards that handle local transactions independently and reconcile cross-shard transfers periodically. Each shard runs its own consensus among geographically

proximate nodes, and cross-shard transactions are verified through merkle proofs and periodic checkpointing to the main chain.

Ten geographic shards each running at 200 TPS provides an effective throughput of 2,000 TPS across the network. Twenty shards provides 4,000 TPS. The sharding architecture scales linearly with geographic distribution, which aligns naturally with Gratia's incentive structure — as more regions gain nodes, more shards can be created.

Geographic sharding is a natural fit for Gratia because the protocol already tracks node locations for Proof of Life and geographic reward weighting. Shard assignment based on physical location is trivial. And because most transactions in a phone-based payment network are local (a person paying a merchant, sending money to a friend, interacting with a local smart contract), the vast majority of transactions stay within their geographic shard without requiring cross-shard coordination.

Long-term scaling path. Beyond geographic sharding, future protocol upgrades (subject to governance approval) could introduce rollup layers, state channels for high-frequency micro-transactions, or other layer-2 scaling solutions. The whitepaper commits to honest, transparent communication about current throughput capabilities rather than making speculative claims about future performance.

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12. Privacy and Security

12.1 Privacy Architecture

Gratia takes a privacy-by-default approach to all data:

- **On-device processing:** All sensor data (GPS, accelerometer, microphone, camera, Wi-Fi, Bluetooth) is processed entirely on-device. Raw sensor data NEVER leaves the phone.
- **Zero-knowledge attestations:** The protocol uses zero-knowledge proofs to attest that a node has valid Proof of Life, is in a valid location, and is running on real hardware — without revealing any underlying data.
- **Unlinkable attestations:** Proof of Life attestations are cryptographically unlinkable between days, preventing tracking of individual nodes over time.

- **User-controlled enhanced features:** Camera and microphone-based attestation features are strictly opt-in. Users who prefer maximum privacy can decline these features and still reach the consensus threshold through other sensors.

12.2 Security Model

- **Hardware-rooted trust:** All cryptographic operations occur within the secure enclave where available. Private keys are generated in hardware and never exist in main memory.
- **Multi-source attestation fraud detection:** Cross-referencing multiple independent sensor sources makes fabricating consistent fake attestations extremely difficult. Spoofing GPS alone is not sufficient — the Wi-Fi, Bluetooth, accelerometer, and charge cycle data must also be consistent.
- **Three-pillar redundancy:** The requirement for simultaneous Proof of Life, staking, and energy expenditure eliminates every known single-vector attack.
- **Quantum resistance consideration:** The secure enclave's hardware-backed key storage provides a natural upgrade path to post-quantum cryptographic algorithms as they mature.

12.3 Proof of Life Hardening

The Proof of Life system faces five categories of spoofing attacks, ranked by sophistication:

1. **Robotic phone farms** — mechanical rigs physically manipulating real phones at scale. Defeated by behavioral analysis detecting mechanical regularity and TEE attestation detecting non-standard hardware interaction.

2. **Rooted phone sensor spoofing** — using Xposed/Magisk frameworks to hook sensor APIs and return fabricated data. Mitigated by TEE attestation (detects root) and cross-day behavioral consistency analysis (fabricated patterns are inconsistent over 30+ day windows).

3. **Replay attacks** — recording legitimate sensor data and replaying it with variations. Countered by requiring live Bluetooth peer discovery (peers change daily and cannot be replayed), tying attestations to current block height, and cross-day consistency checks that detect patterns that are too similar day-to-day.

4. **Human-assisted phone farming** — paying people to carry extra phones with minimal interaction. The hardest to detect because it involves real human motion. Mitigated by requiring behavioral richness (not just unlock counts, but genuine phone usage patterns) and screen-on time correlation.

5. **ARM server emulators** — running Android on ARM cloud servers (AWS Graviton, Ampere) with injected sensor data. Detected by big.LITTLE scheduling fingerprinting (server chips have homogeneous cores), TEE attestation (server VMs fail StrongBox checks), and battery state verification (servers have no battery).

Three defensive layers are deployed against these vectors:

TEE Attestation: Android Play Integrity and iOS DeviceCheck cryptographically prove a device is real, unrooted, and running genuine firmware. While not hard-required (to support custom ROMs and phones without Google services), nodes without TEE attestation face significantly heightened behavioral scrutiny.

Cross-Day Behavioral Analysis: Rather than validating each day independently, the protocol maintains a rolling 30-day behavioral fingerprint for each node. A real human has consistent but naturally varying patterns. The network sees a "behavioral consistency score" attested via zero-knowledge proof — never the underlying data.

Bluetooth Peer Graph Analysis: A network-level defense that detects co-located phone farms by comparing hashed Bluetooth peer sets across nodes. If multiple nodes consistently share the same peer environment, they are flagged for enhanced verification.

12.4 Staking as Security Amplifier

Gratia's staking model is designed as a security mechanism, not an investment vehicle. The core principle: **stake determines how much you can lose, never how much you can earn.** Mining rewards remain flat per minute for every node regardless of stake size.

Three mechanisms launch with the network:

1. **Flat bond** — a fixed security deposit per phone, pegged to 14 days of average mining rewards. Phone farms pay this amount multiplied by every phone they operate, turning scale into a cost multiplier.

2. **Progressive slashing** — graduated penalties that protect honest users while devastating attackers (see Section 5.3).

3. **Fraud reporter incentives** — validators who detect and confirm fraud receive 30% of the slashed stake, creating an active incentive layer for network self-policing.

As the network matures, additional mechanisms activate: mutual staking (trusted nodes vouching for newcomers, with shared slashing risk), geographic stake pooling (heightened scrutiny for regions with detected fraud clusters), and uptime stake decay (reduced stake requirements for long-term honest participants).

12.5 Why Gratia Is Harder to Attack Than Any Existing Blockchain

Every major blockchain's security reduces to a **capital problem.** Bitcoin requires mass-purchasing ASICs and electricity. Ethereum and Solana require mass-purchasing tokens. In every case, the attack scales linearly with money: spend more, get more attack power. A sufficiently funded entity — a nation-state, a hedge fund — can acquire all necessary resources through normal procurement channels. No humans in the loop. Bitcoin 51% attacks have already happened on smaller chains (Ethereum Classic in 2020, Bitcoin Gold in 2018) precisely because the attacker just needed to rent hashpower for a few hours.

Gratia's security reduces to a **human logistics problem.** To attack the network, you don't need more money or more hardware — you need more *people*. Every node requires a living, breathing human carrying a real phone through a real day: unlocking it, interacting with the screen, moving through physical space, encountering different Bluetooth peers, and plugging it in at night. You cannot automate this. You cannot warehouse it. You cannot rent it by the hour.

The practical implications are stark:

Dimension	Traditional Blockchain Attack	Gratia Attack
People required	10-20 engineers	Tens of thousands of carriers
Setup time	Hours to months	Months to years
Can operate from one location	Yes	No — geographic distribution required

Scales with money alone	Yes	No — money buys phones, not reliable humans
Has been done before	Yes (multiple chains)	No equivalent exists
Damage if successful	Steal funds, reverse transactions	Stall blocks (cannot steal or reverse)

At 100,000 honest nodes, reaching 33% network penetration — the threshold for disrupting consensus — requires managing roughly 50,000 phone carriers across multiple countries at an annual cost exceeding $60 million, with an ever-increasing detection risk from behavioral clustering, Bluetooth peer analysis, and geographic anomaly detection. Even then, the attacker can only intermittently stall block production. They cannot steal funds, reverse transactions, or override governance.

The closest real-world analogues to this operation — click farms, survey mills — struggle to maintain 1,000-5,000 reliable participants with constant quality issues, high turnover, and frequent exposure. Nobody has ever covertly managed 50,000 distributed human operators. The attack is theoretically possible but practically absurd — and that is by design.

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13. Wallet Security and Recovery

13.1 The Problem with Existing Wallet Security

Bitcoin and most existing blockchains secure wallets through private keys represented as seed phrases — 12 or 24 random words that users must write down and store safely. If you have the words, you have the money. If you lose the words, your money is gone forever. There is no recovery, no customer support, no reset password option.

This model has caused billions of dollars in permanent losses. An estimated 20% of all Bitcoin ever mined is permanently inaccessible because someone lost their seed phrase. For Gratia's target audience — a farmer in Nigeria, a grandmother in Indonesia, a teenager in Brazil — the seed phrase model is fundamentally incompatible with mass adoption. These users may not have a safe place to store a

piece of paper. They may not be literate. Bitcoin's wallet security was designed by cryptographers for cryptographers. Gratia's wallet security is designed for humans.

13.2 Three-Layer Wallet Protection

Gratia leverages the phone's built-in hardware security and the behavioral data from Proof of Life to create a wallet protection model that is more secure than Bitcoin while being invisible to the user.

Layer 1 — Secure Enclave Key Storage

The private key is generated inside the phone's secure enclave (Apple Secure Element, ARM TrustZone, Android StrongBox) and never leaves it. The key cannot be extracted by the operating system, by malware, by the user, or by Gratia's own app. When a transaction is signed, the signing operation happens inside the secure enclave — the key itself is never exposed to the main processor or memory.

This is fundamentally more secure than Bitcoin's model, where private keys exist in software and can be stolen by malware, keyloggers, or clipboard hijacking. In Gratia, the key exists only in tamper-resistant hardware that is designed to resist even physical extraction attacks.

Layer 2 — Biometric Authorization

Every transaction requires biometric confirmation — fingerprint, face, or device PIN/pattern for phones without biometric hardware. This means that even if someone steals your phone, they cannot move your funds without your fingerprint or face.

Bitcoin has nothing like this at the protocol level. If someone obtains a Bitcoin seed phrase, they do not need the owner's fingerprint, face, or permission. In Gratia, physical possession of the device is not sufficient — biological identity of the owner is required for every value transfer.

Layer 3 — Proof of Life Binding

This is where Gratia does something no other blockchain has ever done. The wallet is cryptographically bound to the owner's Proof of Life behavioral pattern. The protocol knows the behavioral signature of the real owner — their unlock timing patterns, motion signature, interaction cadence, Wi-Fi environment transitions, GPS movement patterns, and daily routine.

If someone steals the phone and somehow bypasses biometric security, their behavioral pattern will not match the owner's. The wallet can detect that a different human is operating the device and lock down automatically, requiring enhanced verification before any funds can move.

13.3 Recovery: Proof of Life Behavioral Matching

The most critical question for any wallet system is: what happens when you lose your phone?

Gratia rejects social recovery (where trusted contacts can collectively restore your wallet) because it introduces a collusion vulnerability — people you trust today may not be trustworthy tomorrow. A bitter divorce, a family dispute, or simple greed could lead trusted contacts to collude and steal funds.

Instead, Gratia uses the one thing that cannot be stolen, shared, or colluded against: the pattern of your daily life.

How recovery works:

Step 1 — Initiate recovery claim. The user obtains a new phone, installs the Gratia app, and initiates a wallet recovery claim by providing their public wallet address (or username/identifier associated with the wallet).

Step 2 — Old wallet is frozen. Immediately upon the recovery claim being filed, the old wallet enters a frozen state. No funds can be moved in or out. If the real owner still has access to their original device, they receive a prominent notification that a recovery claim has been filed against their wallet and can reject it instantly with a single biometric-confirmed tap. A rejected claim ends the process immediately.

Step 3 — Behavioral matching period. If the claim goes uncontested, the new phone begins accumulating Proof of Life data. Over a mandatory waiting period of 7-14 days, the protocol compares the behavioral signature of the new phone against the historical behavioral profile associated with the wallet:

- Unlock timing patterns: Does the new user unlock their phone at similar times of day?
- Wi-Fi environment transitions: Does the new phone connect to the same home Wi-Fi, the same work Wi-Fi, the same regular locations?
- GPS movement patterns: Does the new phone travel the same daily routes — home to work, work to home, the same regular stops?

- Screen interaction cadence: Does the new user interact with their screen in similar rhythms and frequencies?
- Motion signature: Does the accelerometer show similar movement patterns — the same gait, the same commute vibrations, the same daily activity/rest cycle?
- Bluetooth peer environments: Does the new phone encounter the same Bluetooth landscape at home and at work?

Step 4 — Confidence threshold. The protocol computes a behavioral match confidence score. If the score exceeds a conservatively high threshold — meaning the system is confident that the new phone is being used by the same human who owned the wallet — the recovery is approved and the wallet transfers to the new device. If the score falls below the threshold, the recovery is denied and the original wallet is unfrozen.

The confidence threshold is deliberately set high. The system would rather reject a legitimate recovery attempt (the user can simply try for a few more days to accumulate more matching data) than accept a fraudulent one. False negatives are frustrating but recoverable. False positives mean stolen funds. The system errs on the side of caution.

13.4 Why Behavioral Recovery Is Secure

Nobody else lives your life. This simple fact is the foundation of Gratia's recovery security.

A spouse or roommate shares your home Wi-Fi and sleep schedule but goes to a different workplace, connects to different daytime Wi-Fi networks, encounters different Bluetooth peers during the day, and has their own screen interaction patterns and accelerometer gait signature. They produce a partial match at best — not enough to cross the confidence threshold.

A stalker following you visits the same locations but is on their own phone with their own unlock habits, their own interaction cadence, their own gait signature. They'd also need to physically enter your home and workplace to match your Wi-Fi environments. GPS would show two devices in proximity, not one device replacing another.

A thief who steals your phone cannot unlock it without your biometrics. If they factory reset it, the secure enclave is wiped and the keys are gone. They must initiate recovery on a new device, which requires behavioral matching they cannot pass.

A sophisticated technical attack attempting to replay your behavioral patterns on a new device would fail because Proof of Life data is signed by the new phone's secure enclave, which attests that sensor data is coming from real hardware in real time, not from a replay file. Spoofing the secure enclave requires breaking hardware-level security — a nation-state level attack that is not economically rational for wallet theft.

An AI-powered clone of your routine based on observation would get the broad strokes (home location, work location, commute times) but would fail on fine-grained behavioral details — the exact micro-timing of your screen touches, the precise accelerometer signature of your gait, the specific Bluetooth devices your phone encounters in your specific office at your specific desk. These biometric-level signals cannot be replicated through observation alone.

The only scenario that defeats this system is one that defeats every security system ever created: physical coercion, where an attacker forces the owner to unlock the phone and authorize transfers under duress. This is a law enforcement problem, not a protocol design problem.

13.5 Optional Seed Phrase Backup

For users who want an additional layer of security, a traditional seed phrase backup can be generated from within the app settings. This serves as a nuclear recovery option that bypasses behavioral matching entirely.

The seed phrase feature is:

- **Opt-in, not default.** The standard onboarding flow never mentions seed phrases. Most users will never know the feature exists, and they will never need it.
- **Buried in settings.** Generating a seed phrase requires navigating to an advanced security section and confirming with biometric authentication. It is not presented during setup.
- **Accompanied by clear warnings.** Users who generate a seed phrase are informed that anyone who obtains these words can access their funds, and that the seed phrase should be stored securely and never shared.

The seed phrase exists for power users who want maximum control and for extreme edge cases where behavioral matching is not viable (e.g., a user who radically changes their entire daily routine — new home, new city, new job — while simultaneously losing their phone).

13.6 Optional Inheritance Feature

To address the scenario where a wallet owner dies and their funds would otherwise be permanently locked, Gratia offers an optional inheritance feature:

- The user designates a single beneficiary wallet address in their settings.
- If the wallet shows zero Proof of Life activity for a configurable period (default: 365 consecutive days) with no recovery claim filed, the funds automatically transfer to the beneficiary wallet.
- This feature is strictly opt-in and can be modified or revoked at any time by the wallet owner with biometric confirmation.
- The long default period (one year) prevents accidental triggering due to extended hospital stays, incarceration, or other temporary absences.

13.7 Comparison to Existing Wallet Security Models

Feature	Bitcoin	Ethereum (Social Recovery)	Gratia
Key storage	Software (vulnerable to malware)	Software or hardware	Hardware secure enclave (mandatory)
Transaction authorization	Private key only	Private key only	Biometric + secure enclave
Stolen device risk	N/A (keys are in software)	N/A (keys are in software)	Biometric lock + behavioral anomaly detection
Recovery method	Seed phrase or lost forever	Trusted contacts (collusion risk)	Proof of Life behavioral matching (no third-party trust)
Recovery vulnerability	Seed phrase theft	Contact collusion	None identified below nation-state level
Lost key consequence	Funds lost permanently	Contacts can recover	Behavioral matching recovers without third parties
User experience	Must secure 24 words on paper	Must coordinate with trusted	Use your phone normally; plug in and

		contacts	charge
Inheritance	None (funds lost on death)	Possible via social recovery	Optional beneficiary with 365-day dead-man switch

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14. Use Cases

14.1 Digital Cash for the Unbanked

A farmer in sub-Saharan Africa mines GRAT on his $80 Android phone while it charges from a solar panel. He uses NFC tap-to-pay at the local market. No bank account required. No minimum balance. No transaction fees for small amounts.

14.2 Decentralized Gig Economy

A delivery driver's smart contract automatically verifies arrival at the pickup location and delivery destination using location-triggered contracts. Payment releases instantly upon verified delivery — no centralized platform taking a 30% cut.

14.3 Parametric Weather Insurance

Farmers in drought-prone regions purchase smart contract insurance that automatically pays out when the decentralized barometric sensor network detects atmospheric conditions consistent with drought. No insurance adjuster required.

14.4 Event Attendance and Credentialing

Conference attendees receive verifiable attendance NFTs by having their phones NFC-tap a checkpoint, with proximity and location contracts confirming physical presence. These credentials are unforgeable — you had to physically be there.

14.5 Mesh-Resilient Emergency Payments

During a natural disaster that destroys cellular infrastructure, Gratia nodes in the affected area continue processing local transactions via Bluetooth and Wi-Fi Direct mesh. Economic activity doesn't stop because the cell towers went down.

14.6 Decentralized Environmental Monitoring

The millions of barometers, ambient light sensors, and magnetometers in the Gratia network create the world's most granular environmental monitoring system. This data — aggregated, anonymized, and validated by consensus — is available to researchers, governments, and organizations via oracle contracts.

14.7 Incorruptible Polling and Voting

A nonprofit organization needs to survey 100,000 people in a developing country about water infrastructure priorities. Traditional surveys are expensive, slow, and vulnerable to manipulation. Using Gratia's on-chain polling, the organization creates a poll visible to all GRAT holders in the target region. Within days, verified responses flow in — each guaranteed to come from a unique, real human. The results are publicly auditable, tamper-proof, and impossible to manipulate with bots. The same infrastructure could serve political polling, corporate shareholder votes, community decision-making, and any scenario where trustworthy collective human input has value.

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15. App Store Strategy

15.1 The App Store Challenge

Apple's App Store and Google's Play Store have historically restricted or rejected cryptocurrency mining apps, citing battery degradation, excessive resource usage, and user experience concerns. For a protocol that depends entirely on mobile participation, navigating app store policies is not a secondary concern — it is existential.

15.2 Framing: Wallet with Rewards

Gratia's app store positioning is as a **wallet with rewards** rather than a cryptocurrency mining application. This framing is accurate, not misleading:

- In Passive Mode (Proof of Life), the app performs no mining whatsoever. It collects lightweight sensor attestations with zero noticeable battery,

performance, or thermal impact. This is functionally identical to how fitness apps, weather apps, and location-based services already operate.
- In Mining Mode, the phone is plugged in and above 80% battery. There is no battery degradation concern because the device is connected to external power. CPU thermal management is built into the protocol, ensuring the phone never overheats.
- The user experience is that of a wallet that earns rewards when charging — similar to how credit card apps earn cashback or how certain banking apps earn interest on deposits.

15.3 Fallback Distribution

If app store restrictions prevent or delay listing, Gratia will deploy via:

- **Android sideloading:** Android devices can install applications directly from APK files downloaded from the Gratia website, bypassing the Play Store entirely. This is standard practice in many markets.
- **Alternative app stores:** F-Droid (open-source Android store), Samsung Galaxy Store, Amazon Appstore, and Huawei AppGallery provide distribution channels outside Google and Apple's ecosystems.
- **TestFlight (iOS):** Apple's official beta distribution platform supports up to 10,000 testers per app, sufficient for testnet and early adoption phases without requiring full App Store approval.
- **EU alternative app stores:** Under the Digital Markets Act, iOS 17.4+ in the EU supports alternative marketplaces (AltStore PAL, etc.), providing an App Store bypass for European users.

15.4 Long-term Regulatory Trajectory

The regulatory environment for cryptocurrency applications is evolving rapidly. As governments worldwide develop clearer frameworks for digital assets, app store policies are expected to become more accommodating. Gratia's conservative approach — wallet with rewards framing, zero battery impact during passive mode, mining only when charging — positions it well for future policy relaxation. The app's presentation and naming can evolve alongside the regulatory landscape.

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16. Bootstrapping: From Zero to Critical Mass

16.1 The Cold Start Problem

Every blockchain faces the same challenge at launch: why would anyone join a network with no users, no liquidity, and no proven value? The 101st person to install Gratia needs a reason to believe the network is worth their time.

16.2 Intrinsic Value from Block One

Unlike many token launches that depend entirely on speculative demand, GRAT has a measurable intrinsic value from the moment the first block is mined: the energy cost to produce it. Every GRAT requires real electricity consumed by a real phone performing real computational work. This energy-backed floor value exists from day one, independent of exchange listings or market speculation. The first miner can quantify exactly what their GRAT cost to produce, and no rational actor would sell below that cost.

16.3 Early Miner Economics

The emission schedule naturally rewards early participants. With fixed block rewards divided among a small number of initial nodes, early miners earn significantly more GRAT per mining session than later participants. This is not a bonus or a pre-mine — it is the natural mathematics of a fixed reward pool shared among fewer people. Early miners take the biggest risk (participating in an unproven network) and receive the biggest potential reward (high per-node token accumulation before the network scales).

Critically, there is no onboarding delay. A new user downloads the app, plugs in their phone, and starts earning GRAT that same night. The first experience is reward, not waiting. This instant gratification creates immediate engagement — and the progressive trust system ensures that continued mining requires continued honest behavior. Users don't earn the right to mine; they mine immediately and keep the privilege by being real.

16.4 Geographic Launch Strategy

Rather than launching globally and spreading thin, Gratia will target 3-5 specific cities for initial deployment. The goal is to build dense local node clusters where Gratia's unique features — NFC tap-to-pay, Bluetooth mesh networking, proximity contracts — become usable between real people in real daily interactions. When a merchant at a local market accepts GRAT via NFC tap, and a customer pays with GRAT they mined overnight, the network's value proposition becomes tangible and word-of-mouth growth ignites.

Target cities will be selected based on: high smartphone penetration, large unbanked or underbanked population, existing crypto awareness, and a strong community-building potential.

16.5 Community Building Before Launch

The Gratia community begins forming before the first block is mined:

- **Public development:** The protocol is developed openly, with progress visible to anyone who wants to follow along. Transparency builds trust.
- **Waitlist and demo:** A working demo video showing Proof of Life collection, Mining Mode activation, and NFC tap-to-pay creates tangible excitement. A waitlist creates commitment and urgency.
- **Beta testing program:** Early community members receive access to the testnet app, becoming invested participants who evangelize the project from personal experience.
- **Educational content:** Clear, accessible explanations of why existing blockchains have failed the average person and how Gratia is different. The narrative — "your phone works for you while you sleep" — is simple enough for anyone to understand and compelling enough to share.

16.6 Network Effects

Gratia's utility increases non-linearly with adoption:

- At 100 nodes: proof of concept, core team testing.
- At 10,000 nodes: testnet validation, community formation.
- At 100,000 nodes: viable regional payment network in target cities, NFC commerce becomes practical.
- At 1,000,000 nodes: robust global network, meaningful environmental sensor data, on-chain polling gains commercial value, geographic sharding activates.

- At 10,000,000+ nodes: the most decentralized blockchain in existence, with more nodes than Bitcoin, Ethereum, and Solana combined.

Each milestone unlocks new utility that attracts the next wave of participants, creating a self-reinforcing adoption cycle.

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17. Regulatory Positioning

17.1 GRAT as a Commodity

Gratia believes GRAT exhibits the characteristics of a commodity rather than a security. Under the Howey Test — the primary legal framework used in the United States to determine whether an asset is a security — an asset is a security if it involves (1) an investment of money (2) in a common enterprise (3) with an expectation of profit (4) derived from the efforts of others.

GRAT does not satisfy these criteria:

- **Effort of the holder, not others.** GRAT is earned through the miner's own effort — their phone's energy expenditure, their daily Proof of Life attestation, their staked commitment. Mining rewards are directly proportional to the individual's own participation, not dependent on the efforts of a central team or third party.
- **Intrinsic utility beyond investment.** GRAT has immediate utility as digital cash (peer-to-peer payments, NFC tap-to-pay), smart contract fuel, on-chain polling infrastructure, and governance participation. It is not purchased solely with an expectation of profit.
- **Decentralized production.** No single entity controls the creation or distribution of GRAT. Tokens are produced by a distributed network of independent phone nodes with no central coordinator. This mirrors the commodity classification applied to Bitcoin by the U.S. Commodity Futures Trading Commission (CFTC) and to Ethereum by both the CFTC and SEC.
- **Energy-backed creation.** Like Bitcoin, each GRAT token requires measurable energy expenditure to produce, giving it commodity-like production characteristics analogous to mining a physical resource.

17.2 Regulatory Engagement

Gratia will proactively engage with regulators rather than attempting to avoid regulatory scrutiny. The project's transparent fair launch model (no private pre-sale, no discounted investor tokens, open-source code, publicly visible development) is designed to demonstrate good faith. The team will seek legal counsel in key jurisdictions and pursue regulatory clarity before mainnet launch rather than after.

17.3 Global Regulatory Considerations

Cryptocurrency regulation varies dramatically by jurisdiction. Gratia's global nature requires awareness of and compliance with:

- **United States:** Commodity classification aligns with CFTC precedent for decentralized, mined tokens. No securities offering is planned.
- **European Union:** The Markets in Crypto-Assets (MiCA) framework provides clear guidelines for utility tokens and crypto-asset service providers.
- **Developing markets:** Many of Gratia's target regions (Sub-Saharan Africa, Southeast Asia, Latin America) are actively developing crypto-friendly regulatory frameworks. Gratia's mission of financial inclusion aligns with the stated goals of many of these regulatory initiatives.

This whitepaper does not constitute legal advice and makes no definitive claims about GRAT's regulatory classification in any jurisdiction. The project will engage qualified legal counsel in each relevant jurisdiction.

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18. Technical Architecture: Cross-Platform via Shared Core

Platform Support

Gratia runs on both Android and iOS. The application architecture ensures that both platforms are fully supported with approximately 80% shared code.

Shared Rust Core

The protocol's entire computational engine — consensus, Proof of Life validation, wallet operations, zero-knowledge proofs, networking, state management, and

transaction processing — is implemented in Rust. Rust was selected because it compiles to highly optimized native ARM64 binaries on both Android and iOS, has no garbage collector (critical for predictable consensus performance within 3-5 second block windows), and automatically leverages ARM hardware cryptographic accelerators (AES, SHA) built into every modern mobile chipset.

This single Rust codebase cross-compiles to Android, iOS, and server targets. There is no separate implementation per platform — the same logic runs on both.

Thin Native Layers

On each platform, a thin native layer handles two responsibilities: reading phone sensors (which require platform-specific APIs) and rendering the user interface (which uses each platform's native UI toolkit for the best experience).

On Android, this layer is written in Kotlin using Android Sensor APIs and Jetpack Compose for the interface. On iOS, it is written in Swift using Core Motion, Core Location, Core Bluetooth, and SwiftUI for the interface.

These native layers communicate with the shared Rust core through Mozilla's UniFFI framework, which auto-generates Kotlin bindings (Android) and Swift bindings (iOS) from the same Rust source code. This approach is battle-tested — Mozilla uses it for Firefox on both platforms.

Zero-Knowledge Transaction Privacy

Transactions on Gratia are transparent by default — sender, receiver, and amount are visible on chain, similar to Bitcoin. However, users can opt into shielded transactions on a per-transaction basis. Shielded transactions use Bulletproofs and Pedersen commitments to prove transaction validity without revealing the amount or parties involved. The user selects shielded mode with a single tap in the app. Proof generation takes 2-5 seconds on a typical ARM chipset and is designed to run during Mining Mode (plugged in, above 80%) when the phone has power to spare. Users can queue a shielded transaction while unplugged and the proof generates automatically the next time mining conditions are met.

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19. Implementation Status

Gratia is not a concept — it is a working protocol running on real smartphones. The following has been built, tested, and demonstrated on production hardware.

19.1 Proven on Real Hardware

The protocol has been deployed and tested on two Android smartphones:

- **Samsung Galaxy A06** — a $90 budget phone (Mediatek Helio G85, 4GB RAM)
- **Samsung Galaxy S25** — a flagship phone (Snapdragon 8 Elite, 12GB RAM)

Both devices run the identical Gratia app compiled from the same Rust core. The budget phone performs identically to the flagship for all protocol operations — block production, transaction processing, PoL attestation, consensus participation. This validates the design goal: any $50+ phone manufactured after 2018 can fully participate.

19.2 Test Coverage

The protocol is backed by 705 automated tests across the full workspace:

Category	Tests	What They Validate
Core protocol types & crypto	86	Ed25519, SHA-256, transaction serialization, address derivation
Proof of Life validation	102	All 8 PoL parameters, behavioral analysis, suspicious pattern detection
Staking & overflow pool	54	Caps, overflow distribution, cooldown, slashing
Consensus & VRF	61	Committee selection, block production, VRF proofs, graduated scaling
Wallet & transactions	45	Key generation, signing, verification, shielded transactions
Network & gossip	34	Gossipsub, deduplication, peer discovery, sync protocol
Phone farm attack simulation	18	Scripted unlock detection, narrow spread rejection, single-BT-environment rejection
Sybil resistance	13	Stake caps block Sybil advantage, whale power capped, slash/ban enforcement

Behavioral spoofing	10	Emulator TEE penalties, presence score boundaries, consistency bonuses
Stake manipulation	10	Zero-stake rejection, exact-cap overflow, yield distribution, eligibility loss
GratiaScript compiler	86	Lexer tokenization, parser AST construction, WASM code generation, type checking
WASM interpreter	17	Instruction execution, host function dispatch, gas metering, control flow
Smart contract e2e	19	Compile → deploy → execute pipeline, template contracts, error handling
Governance	53	Proposal lifecycle, voting mechanics, poll creation, one-phone-one-vote
GratiaVM & sandbox	49	Contract deployment, gas accounting, permissions, resource limits

19.3 Performance Metrics

Metric	Value
Block time	4 seconds
Base throughput	128 TPS (single chain)
With geographic sharding	512-2,560 TPS (4-20 shards)
Transaction size	~200 bytes
Max block size	256 KB (512 transactions)
On-chain state per phone	< 1 KB (current), 2-5 GB target at scale
Consensus finality	14/21 validator signatures (67%)
App startup to first block	~15 seconds

19.4 Security Validation

The three-pillar consensus security model (Proof of Life + Staking + Energy) has been validated through simulation:

- **Phone farms rejected**: Scripted unlock patterns fail the 6-hour spread requirement. Single Bluetooth environments are caught. Missing sensor data (no motion, no orientation, no GPS) fails validation immediately.
- **Sybil attacks blocked**: Stake cap ensures 10 Sybil nodes have identical consensus power to 1 legitimate node. No advantage from duplication.
- **Emulators detected**: TEE attestation failure incurs a -15 point penalty on presence score, dropping emulators below the 40-point consensus threshold.
- **Whale power capped**: A node staking 100x the cap has the same consensus power as a minimum staker. Excess stake earns yield but cannot buy influence.
- **Transaction forgery prevented**: All incoming transactions are verified via Ed25519 signature + hash check before any balance is credited.

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20. Development Roadmap

Phase 1 — Foundation ✅ COMPLETE

- ✅ Shared Rust core library: 13 crates (core, pol, wallet, zk, staking, consensus, network, state, governance, vm, ffi, gratiascript, interpreter)
- ✅ 705 passing tests with zero warnings across the full workspace
- ✅ GratiaScript compiler: TypeScript-derived syntax → WASM bytecode with mobile-native opcodes (@location, @proximity, @presence, @sensor)
- ✅ Pure-Rust WASM interpreter: executes smart contracts without wasmer/LLVM dependency — compiles for any ARM64 target
- ✅ GratiaScript type checker: compile-time type resolution, scoping, immutability enforcement
- ✅ One-phone-one-vote governance: proposals, polls, voting wired end-to-end from Rust through FFI to Android UI
- ✅ Proof of Life parameter collection via Android sensor APIs (GPS, accelerometer, Bluetooth, Wi-Fi, battery, barometer, magnetometer, ambient light)
- ✅ Rust-side PoL validation with all 8 daily parameters enforced
- ✅ Mining Mode activation (plugged in + battery ≥ 80% + valid PoL + minimum stake)
- ✅ Ed25519 wallet with key persistence, transaction signing, and signature verification on all incoming transactions
- ✅ Bulletproofs integration for zero-knowledge PoL attestations

- ✅ Phone-to-phone consensus demonstrated on 2 real devices (Samsung Galaxy A06 and S25) via libp2p QUIC + mDNS
- ✅ NFC tap-to-transact (HCE service + reader mode)
- ✅ QR code scanning for wallet address transfer
- ✅ Jetpack Compose UI: Wallet (send/receive/history), Mining (status/PoL/staking), Network (peers/consensus), Governance (proposals/polls/voting), Settings
- ✅ Cross-compiled to Android ARM64 via NDK, UniFFI Kotlin bindings auto-generated

Phase 2 — Testnet ✅ COMPLETE

- ✅ VRF-based block production with 4-second slot timer, running on real phones
- ✅ On-chain state management with account balances, nonces, and file-based persistence across app restarts
- ✅ Transaction mempool: verified transactions flow from gossip → mempool → block inclusion (real on-chain TPS)
- ✅ Ed25519 signature + hash verification on all received transactions; balance and nonce checks for known accounts
- ✅ Block sync protocol: gossip-based catchup broadcasts recent blocks on peer connect
- ✅ Staking UI: stake/unstake from the Mining tab, overflow pool visible
- ✅ Governance: create proposals (14-day discussion + 7-day voting), create polls, one-phone-one-vote
- ✅ GratiaVM smart contract engine: MockRuntime with 3 demo contracts demonstrating @location, @proximity, @presence opcodes; ARM-calibrated gas metering; sandboxing with memory/time/call-depth limits
- ✅ Block explorer web app (HTML/CSS/JS) with live HTTP API serving real chain data from phones
- ✅ Network simulation tests: 51 tests covering phone farm detection, Sybil resistance, behavioral spoofing, emulator TEE penalties, stake manipulation edge cases

Phase 3 — Public Testnet (Next)

- Bootstrap server for internet-wide peer discovery (beyond same-LAN mDNS)
- Full WASM execution via wasmer (currently MockRuntime with native handlers)

- GratiaScript compiler: TypeScript-derived language → WASM bytecode
- RocksDB persistent state storage (replacing file-based InMemoryStore serialization)
- iOS app via Swift sensor layer + same Rust core (UniFFI Swift bindings)
- Dynamic transaction fee system based on congestion and computation
- Security audit of Rust core, ZK proofs, and consensus mechanism

Phase 4 — Mainnet (Months 13-18)

- Mainnet launch with genesis block mined on real phones under same rules as everyone
- Android and iOS apps in respective app stores (positioned as "wallet with rewards")
- Geographic sharding activation (4-20 shards, up to 2,560 TPS)
- Bluetooth/Wi-Fi Direct mesh layer for offline transactions
- Oracle layer for aggregated environmental sensor data
- Bug bounty program

Phase 5 — Scale (Year 2+)

- Target: 1 million active mining nodes
- Cross-chain bridges to Bitcoin and Ethereum ecosystems
- Developer ecosystem grants and GratiaScript SDK
- Enterprise and government partnerships for verified polling and environmental data
- Localized deployment in unbanked regions
- DAO governance transition

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21. Team and Governance

[To be developed — founder background, advisory board, organizational structure]

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22. Conclusion

Gratia does not ask people to buy expensive hardware to participate in the future of finance. It does not require capital deposits that exclude the majority of the world's population. It does not consume the energy output of small countries to validate transactions.

Gratia asks only that you have a phone — Android or iPhone, flagship or budget, new or a few years old — and that you plug it in.

In doing so, you become part of a global network that is more decentralized than Bitcoin (because its nodes are more numerous and geographically distributed), more accessible than Ethereum (because there is no financial barrier to participation), and more useful than both (because its smart contracts can interact with the physical world through the sensors in your pocket).

The protocol's security does not rest on who has the most money or the most hardware. It rests on the most fundamental proof imaginable: that you are alive, that you are here, and that you choose to participate. Your daily life — the way you pick up your phone, carry it with you, charge it at night — is all the proof the network needs.

The word "gratia" means grace, gratitude, and freedom. The network embodies all three: the grace of elegant design, gratitude for every participant who strengthens it, and freedom for every person who can now access a global financial system with nothing more than the phone in their hand.

One phone. One voice. One network for everyone.

This document describes a working protocol (v0.5) that has been implemented and tested on real hardware. All parameters, thresholds, and mechanisms described herein have been validated through 705 automated tests and real-device deployment. Specifications remain subject to revision based on ongoing research, security analysis, and community input.

Gratia Project Contact: [TBD] **Website:** gratia.io **Ticker:** GRAT **Smallest Unit:** 1 Lux (1 GRAT = 1,000,000 Lux)